ÖZYEĞİN UNIVERSITY
Math 104 – Spring 2018

Final Exam

Seat Number: .....

Name: .....

Number: .....

- Please read all directions and sign the honor code below.
- The exam will be 120 minutes long and consists of 9 questions in 8 pages. Make sure you have all the pages.
- Be sure to write your exam seat number, full name, and student number above.
- Calculators and other electronic devices such as smart phones are not allowed.
- Leaving the exam room temporarily during the exam is not allowed.
- You cannot leave the exam during the first 35 minutes.
- Please raise your hand quietly if you have any questions for the instructors or for the proctors.
- Answers given without supporting work will receive no credit.

Honor Code: I pledge under my honor that I **did not give or receive any unauthorized assistance** in this work.

Signed:.....

Question	Max. Points	Score
1	10	
2	10	
3	10	
4	10	
5	8	
6	10	
7	16	
8	12	
9	14	
Total	100	

Q1) [2×5 = 10pts] **(The Chain Rule)**

(i) Find $\frac{\partial w}{\partial q}$ when $p = \frac{\pi}{6}$, $q = 0$ if $w = \sin x \sin(yz)$, $x = 3p - 2q$, $y = p + q$, and $z = \frac{\pi}{p}$.

$w = \sin x \sin(yz)$ $\quad x = 3p - 2q$, $y = p + q$ $\quad z = \frac{\pi}{p}$

For $(p,q) = (\frac{\pi}{6}, 0)$, $(x,y,z) = (\frac{\pi}{2}, \frac{\pi}{6}, 6)$

$$\frac{\partial w}{\partial q} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial q} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial q} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial q}$$

$$= \cos x \sin(yz)(-2) + \sin x \, z \cos(yz) \cdot 1 + 0$$

$$\left.\frac{\partial w}{\partial q}\right|_{(\frac{\pi}{6}, 0)} = \underbrace{\cos\frac{\pi}{2}}_{0} \underbrace{\sin\pi}_{0}(-2) + \sin\frac{\pi}{2} \cdot 6 \cdot \cos\pi = -6$$

(ii) Assume that $w = f(x^2y, \frac{x}{y})$, $\frac{\partial f}{\partial r}(r,s) = rs$ and $\frac{\partial f}{\partial s}(r,s) = \frac{r^2}{2}$. Find $\frac{\partial w}{\partial x}$.

Let $r = x^2y$ and $s = \frac{x}{y}$. Then $w = f(r,s)$

$$\frac{\partial w}{\partial x} = \frac{\partial f}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial f}{\partial s}\frac{\partial s}{\partial x}$$

$$= rs \cdot (2xy) + \frac{r^2}{2}\frac{1}{y}$$

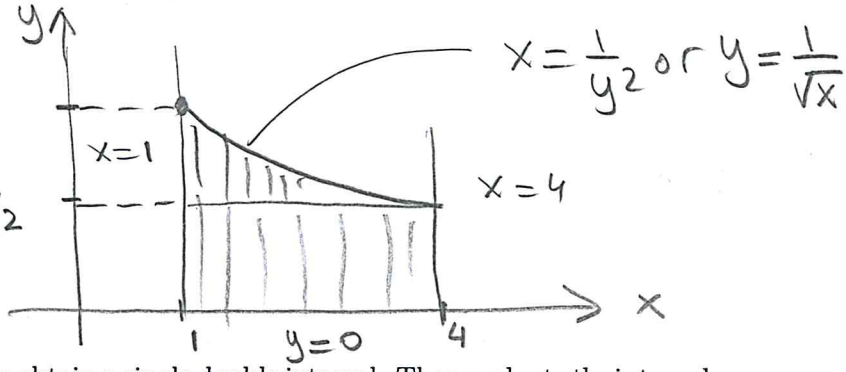
$$\frac{\partial w}{\partial x} = x^2\cancel{y}\frac{x}{\cancel{y}}(2xy) + \frac{x^4y^{\cancel{2}}}{2}\frac{1}{\cancel{y}} = 2x^4y + \frac{1}{2}x^4y$$

$$= \frac{5}{4}x^4y$$

Q2) [4 + 6 = 10pts] **(Double Integrals over General Regions)** For the following double integrals

$$I = \int_0^{1/2}\int_1^4 \cos(\tfrac{\pi}{2}\sqrt{x})dxdy + \int_{1/2}^1\int_1^{1/y^2} \cos(\tfrac{\pi}{2}\sqrt{x})dxdy,$$

(i) Sketch the region of integration.



(ii) Reverse the order of integration to obtain a single double integral. Then evaluate the integral.

$$I = \int_{x=1}^{4}\int_{y=0}^{\frac{1}{\sqrt{x}}} \cos\left(\frac{\pi}{2}\sqrt{x}\right) dy\,dx = \int_1^4 \cos\left(\frac{\pi}{2}\sqrt{x}\right)[y]_0^{\frac{1}{\sqrt{x}}} dx$$

$$= \int_1^4 \frac{1}{\sqrt{x}}\cos\left(\frac{\pi}{2}\sqrt{x}\right)dx = \frac{4}{\pi}\int_{\pi/2}^{\pi}\cos u\,du$$

$$u = \frac{\pi}{2}\sqrt{x} \Rightarrow du = \frac{\pi}{4}\frac{dy}{\sqrt{x}} \qquad = \frac{4}{\pi}(\sin u)\Big|_{\pi/2}^{\pi} = -\frac{4}{\pi}$$

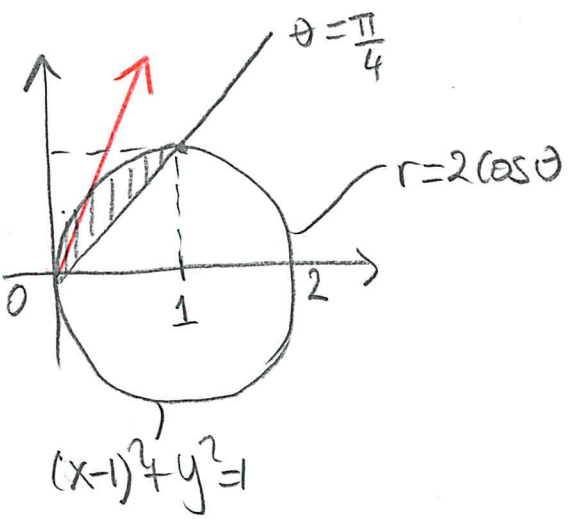
Q3) [10pts] **(Double Integrals in Polar Form)** Change the following Cartesian integral into an equivalent polar integral. Then evaluate the polar integral.

$$J = \int_0^1\int_x^{\sqrt{1-(x-1)^2}} \frac{1}{\sqrt{x^2+y^2}}dydx$$

$x = r\cos\theta \quad y = r\sin\theta \quad dA = r\,dr\,d\theta$

$y = x \Rightarrow \theta = \frac{\pi}{4}$; $x^2+y^2 = r^2$

$y = \sqrt{1-(x-1)^2} \Rightarrow x^2+y^2-2x = 0 \Rightarrow r = 2\cos\theta$



$$J = \int_{\pi/4}^{\pi/2}\int_0^{2\cos\theta} \frac{1}{\cancel{r}}\cancel{r}\,dr\,d\theta = \int_{\pi/4}^{\pi/2} 2\cos\theta\,d\theta$$

$$J = 2\sin\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = 2\left(1-\frac{1}{\sqrt{2}}\right)$$

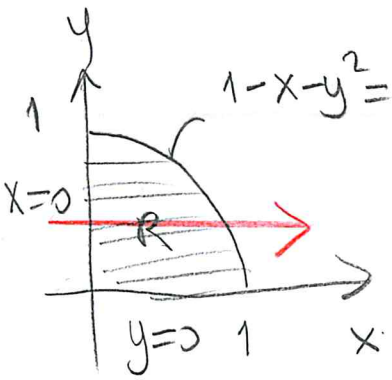
x^2

$(x>0, y>0, z>0)$

Q4) [10pts] **(Triple Integrals in Rectangular Coordinates)** Use triple integrals to find the volume of the region in the first octant bounded by the coordinate planes and the surface $z = 1 - x - y^2$.

$$V = \iiint_D dV = \iint_R \left(\int_{z=0}^{1-x-y^2} dz \right) dA = \iint_R (1-x-y^2)\, dA$$

For $z=0$, $1-x-y^2=0$, $x=0$ and $y=0$ are the boundary of R:



$$V = \int_{y=0}^{1} \int_{x=0}^{1-y^2} (1-x-y^2)\, dx\, dy$$

$$V = \int_0^1 \left(x - \frac{x^2}{2} - xy^2\right)\Big|_0^{1-y^2} dy = \int_0^1 (1-y^2)\left[1 - \frac{1}{2}(1-y^2) - y^2\right] dy$$

$$V = \frac{1}{2}\int_0^1 (1-y^2)^2\, dy = \frac{4}{15}$$

Q5) [4+4=8pts] **(Infinite Series)** Find the series' sum if the series converges or show that the series diverges.

(i) $\displaystyle\sum_{n=1}^{\infty} \left(\tan\left(\frac{\pi}{4n}\right) - \tan\left(\frac{\pi}{4(n+1)}\right) \right)$, (ii) $\displaystyle\frac{1}{3} - \frac{1}{2} + \frac{3}{4} - \frac{9}{8} + \cdots + (-1)^{n-1}\frac{1}{3}\left(\frac{3}{2}\right)^{n-1} + \cdots$

i) n^{th} partial sum S_n:

$S_n = a_1 + a_2 + \cdots + a_n$

$S_n = \left(\tan\frac{\pi}{4} - \cancel{\tan\frac{\pi}{8}}\right) + \left(\cancel{\tan\frac{\pi}{8}} - \cancel{\tan\frac{\pi}{12}}\right) + \cancel{\cdots} + \left(\cancel{\tan\frac{\pi}{4(n-1)}} - \cancel{\tan\frac{\pi}{4n}}\right) + \left(\cancel{\tan\frac{\pi}{4n}} - \tan\frac{4}{4n+1}\right)$

$S_n = \tan\frac{\pi}{4} - \tan\frac{4}{4n+1}$

$\lim_{n\to\infty} S_n = \tan\frac{\pi}{4} - \tan 0 =$

$= 1 = \sum a_n$

ii) $\displaystyle\sum_{n=1}^{\infty} (-1)^{n-1}\frac{1}{3}\left(\frac{3}{2}\right)^{n-1}$ is a geometric series with $a = \frac{1}{3}$ and $r = -\frac{3}{2}$.

Since $|r| = \left|-\frac{3}{2}\right| > 1$, the GS diverges

Q6) [6+4=10pts] **(Integral Test)** *(i)* Use the Integral Test to determine if the series $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ converges or diverges. Be sure to check that the conditions of the Integral Test are satisfied.

Let $f(x) = \frac{1}{x \ln x}$ with $f(n) = a_n$. Since f is positive for $x \geq 2$, continuous and decreasing function, we can use the Integral Test:

$$\int_2^{\infty} f(x)\,dx = \int_2^{\infty} \frac{dx}{x \ln x} \underset{u = \ln x}{=} \int_{\ln 2}^{\infty} \frac{du}{u} = \ln u \Big|_{\ln 2}^{\infty} = \infty.$$

By the Integral T., both the Improper Integral $\int_2^{\infty} \frac{dx}{x \ln x}$ and the series $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ diverges.

(ii) Does the series $\sum_{n=2}^{\infty} (-1)^n \overbrace{\frac{1}{n \ln n}}^{u_n}$ converge or diverge? Alternating Series

By the Alternating Series Test

a) $u_n = \frac{1}{n \ln n} > 0$ for $n \geq 2$

b) $u_n = \frac{1}{n \ln n} > \frac{1}{(n+1)\ln(n+1)} = u_{n+1}$

c) $\lim_{n \to \infty} \frac{1}{n \ln n} = 0$; the AS $\sum_{n=2}^{\infty} (-1)^n \frac{1}{n \ln x}$ converges

Q7) [4×4=16pts] **(Infinite Series)** Use any method to determine if the series converges or diverges. Give reasons for you answer.

(i) $\sum_{n=1}^{\infty} \frac{n}{n+1} = \sum a_n$, Let $a_n = \frac{n}{n+1}$.

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{n+1} = 1 \neq 0$. By the nth term test $\sum a_n = \sum \frac{n}{n+1}$ diverges

(ii) $\sum_{n=1}^{\infty} (1-\frac{2}{n})^{n^2} n$

Apply the Root Test: $\sqrt[n]{a_n} = \left[(1-\frac{2}{n})^{n^2} n\right]^{\frac{1}{n}} = (1-\frac{2}{n})^n \sqrt{n}$

$\lim_{n\to\infty} \sqrt[n]{a_n} = \lim_{n\to\infty} (1-\frac{2}{n})^n \sqrt{n} = e^{-2} \cdot 1 = \frac{1}{e^2} < 1$, that is $\sum a_n$ converges

(iii) $\sum_{n=1}^{\infty} \frac{n! 3^{2n}}{n^2(n+2)!}$

Apply the Ratio Test

$\frac{a_{n+1}}{a_n} = \frac{(n+1)!\, 3^{2(n+1)}}{(n+1)^2 (n+3)!} \cdot \frac{n^2 (n+2)!}{n!\, 3^{2n}} = \frac{\cancel{(n+1)}}{n+3} \frac{n^2\, 3^2}{(n+1)^{\cancel{2}}} \to 9 > 1$

By the Ratio Test, $\sum a_n$ diverges

(iv) $\sum_{n=2}^{\infty} \frac{(\ln n)^2}{n^3}$ (Hint: Use $\ln n < n^c$ for large n, $c > 0$)

$a_n = \frac{(\ln n)^2}{n^3} < \frac{(n^c)^2}{n^3} = \frac{n^{2c}}{n^3} = \frac{n}{n^3} = \frac{1}{n^2} = b_n$ for $c = \frac{1}{2}$.

By the CT, $\sum a_n = \sum \frac{(\ln n)^2}{n^3}$ converges

since $\sum b_n = \sum \frac{1}{n^2}$ converges.

(p-series with $p = 2 > 1$)

Q8) [3×4=12pts] **(Power Series)** *(i)* Find the series' radius and interval of convergence. *(ii)* For what values of x does the series converge *(a)* absolutely, *(b)* conditionally? *(iii)* Find the sum of the series as a function of x in the interval of convergence.

$$\sum a_n = \sum_{n=0}^{\infty} \frac{(4x-6)^{n+1}}{2(n+1)}$$

i) Use the Ratio Test

$$\frac{|a_{n+1}|}{|a_n|} = \frac{|4x-6|^{n+2}}{\cancel{2}(n+3)} \frac{\cancel{2}(n+1)}{|4x-6|^{n+1}} \Rightarrow |4x-6|<1 \quad \text{with } R=\frac{1}{4}$$

$$\frac{5}{4}<x<\frac{7}{4}$$

Endpoints:

$x=\frac{5}{4}$: $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{2(n+1)}$ Alternetin Series converges by the Alternating Series Test

$x=\frac{7}{4}$: $\sum_{n=0}^{\infty} \frac{1}{2(n+1)}$ diverges by comparing with the Harmonic series $\sum \frac{1}{n}$

Interval of convergence: $\frac{5}{4} \le x < \frac{7}{4}$

ii) The series converges absolutely in $\frac{5}{4}<x<\frac{7}{4}$ and converges conditionally at $x=\frac{5}{4}$

iii) $f(x) \overset{?}{=} \sum_{n=0}^{\infty} \frac{(4x-6)^{n+1}}{2(n+1)}$ for $|4x-6|<1$

Differentiating the series term-by-term, we have

$$f'(x) = \sum_{n=0}^{\infty} \frac{4}{2} \underbrace{(4x-6)^n}_{\text{geometric series}} = 2\frac{1}{1-(4x-6)} = \frac{2}{7-4x} \text{ for } |4x-6|<1$$

Integrating $f'(x)$ we get

$$f(x) = \int f'(x)dx = 2\int \frac{dx}{7-4x} = -\frac{2}{4}\ln(7-4x)+C; \quad f\left(\frac{6}{4}\right)=0 \Rightarrow C=0$$

$$f(x) = -\frac{1}{2}\ln(7-4x) = \sum_{n=0}^{\infty} \frac{(4x-6)^{n+1}}{2(n+1)} \text{ for } |4x-6|<1$$

Q9) [4+3+3+4=14pts] **(Taylor and Maclaurin Series)** *(i)* Find the Maclaurin series for the function $f(x) = \dfrac{1}{1+x}$.

$f(x) = (1+x)^{-1}$ $\quad f'(x) = -(1+x)^{-2}$ $\quad f''(x) = 2(1+x)^{-3}$

$f^{(n)}(x) = (-1)^n n! \dfrac{1}{(1+x)^{n+1}}$; $\quad f^{(n)}(0) = (-1)^n n!$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} (-1)^n \frac{n!}{n!} x^n = \sum_{n=0}^{\infty} (-x)^n$$ geometric series

$$= 1 - x + x^2 - x^3 + \cdots + (-x)^{n-1} + \cdots$$

(ii) Use the series in part *(i)* to find the value of f at $x = 1/2$.

The sum of the GS with $|r| = |-x| < 1$ is

$$1 - x + \cdots + (-x)^{n-1} + \cdots = \frac{1}{1-(-x)} = \frac{1}{1+x} = f(x); \quad f\left(\tfrac{1}{2}\right) = \frac{2}{3}$$

(iii) Can you use the series in part *(i)* to find the value of f at $x = 2$? Explain.

We cannot use the Maclaurin series (also the GS) to find f at $x=0$ since the series diverges for $|x| = 2 > 1$

(iv) Write the Taylor polynomials of orders 2 and 3 generated by $f(x) = \dfrac{1}{1+x}$ at $a = 0$.

$P_2(x) = 1 - x + x^2$

$P_3(x) = 1 - x + x^2 - x^3$